CDA 106
Numerical Methods for Data Science

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QR decomposition

$$A = QR$$

Here $Q :=$ Orthogonal Matrix

$R :=$ Upper triangular matrix.

Orthogonal Matrix $\longrightarrow Q$

if $QQ^T = I = Q^T Q$

This is helpful in solving systems $Ax = b$.

$$Q_{n\times n} = \begin{bmatrix} q_{11} & q_{12} & \cdots & q_{1n} \\ q_{21} & q_{22} & \cdots & q_{2n} \\ \vdots & & & \\ q_{n1} & q_{n2} & \cdots & q_{nn} \end{bmatrix} = \begin{bmatrix} q_1 \\ \hline q_2 \\ \hline \vdots \\ \hline q_n \end{bmatrix}_{n\times 1} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix}_{n\times 1}$$

wh. $q_i^T = \begin{bmatrix} q_{i1} \\ q_{i2} \\ \vdots \\ q_{in} \end{bmatrix}$ ith column vector.

$$Q^T = \begin{bmatrix} q_1^T & | & \cdots & | & q_n^T \end{bmatrix}_{1\times n} = \begin{bmatrix} q_1^T & q_2^T & \cdots & q_n^T \end{bmatrix}_{1\times n}$$

$$Q^T = \begin{bmatrix} q_{11} & q_{21} & \cdots & q_{n1} \\ q_{12} & q_{22} & \cdots & q_{n2} \\ \vdots & \vdots & & \vdots \\ q_{1n} & q_{2n} & & q_{nn} \end{bmatrix}$$

$q_1^T \quad q_2^T \quad q_n^T$

$$QQ^T = \begin{bmatrix} q_1 q_1^T & q_1 q_2^T & \cdots & q_1 q_n^T \\ q_2 q_1^T & q_2 q_2^T & & \\ \vdots & & & \\ & & & q_n q_n^T \end{bmatrix}_{n\times n}$$

$$q_i \cdot q_j^T = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

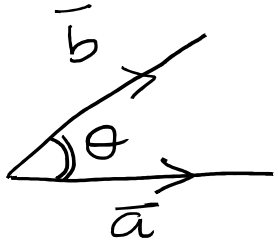
$$\begin{bmatrix} \text{———} \end{bmatrix} \begin{bmatrix} | \\ | \end{bmatrix}$$

Vector product: (Dot)

$\bar{a} = (a_1, a_2, a_3)$

$\bar{b} = (b_1, b_2, b_3)$

3D



$\bar{a} \cdot \bar{b} = |a|\,|b| \cos\theta$

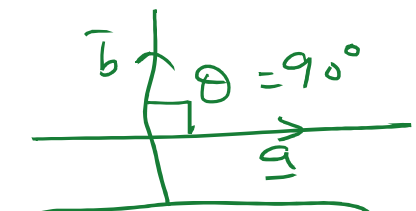
orthogonal

$\bar{a}$ is perpendicular to $\bar{b}$ if $\bar{a} \cdot \bar{b} = 0$

because $\bar{a} \cdot \bar{b} = 0 \Rightarrow |a||b|\cos\theta = 0$

$|a||b| \neq 0 \Rightarrow \cos\theta = 0$

$\Rightarrow \theta = \pi/2$



$\boxed{\bar{a} \cdot \bar{b} = 0}$

$\bar{a} \cdot \bar{b} = 0 \Rightarrow a_1 b_1 + a_2 b_2 + a_3 b_3 = 0$

n dimension

In higher dimension,

$\bar{a} \perp \bar{b}$ ($\bar{a}$ is orthogonal to $\bar{b}$) if

$\bar{a} = (a_1, a_2, \ldots, a_n)$ $\quad \bar{b} = (b_1, b_2, \ldots, b_n)$

$\bar{a} \perp \bar{b}$ if $\bar{a} \cdot \bar{b} = 0$

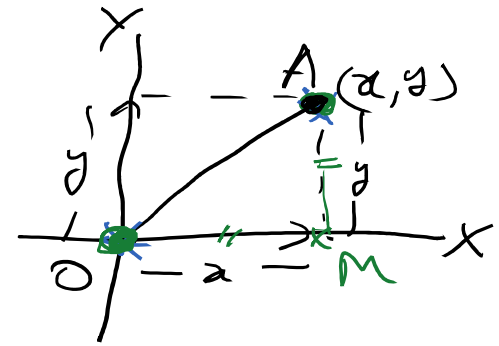
$\qquad\qquad\qquad\qquad \parallel$

or $\boxed{a_1 b_1 + a_2 b_2 + \cdots + a_n b_n = 0}$ ✓

Suppose $\bar{a} \cdot \bar{a} = a_1^2 + a_2^2 + a_3^3$ (magnitude^2)

$= |a|^2$ ✓

Distance



① $OA^2 = x^2 + y^2$

$$OA = \underbrace{\sqrt{x^2+y^2}}$$ (Euclidean)

There can be other notion of distances as well for measuring distance between O & A.

② $OA = OM + MA$
$= |x| + |y|$

③ $OA = \text{Max}\{|x|, |y|\}$

Mathematically, in higher dimension, we denote this (notion of distance)

Norm $\|\cdot\|_X$

Inner product

$V(\mathbb{R})$ $\quad (V = \mathbb{R}^n, \mathbb{F} = \mathbb{R})$

$a, b \in V = \mathbb{R}^n$

Defn $\underbrace{\langle a, b \rangle}_{\text{inner product of } a \,\&\, b} = a^T b = [a_1 \; a_2 \; \cdots \; a_n] \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$

$= a_1 b_1 + a_2 b_2 + a_n b_n$

$a = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^n$

(Please keep in mind)

$$\langle a, a \rangle = a^T a = [a_1 \; a_2 \; \cdots \; a_n] \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

$$= a_1^2 + a_2^2 + \cdots + a_n^2$$

$$= \|a\|^2$$

For orthogonality of a & b, i.e. $a \perp b$ if $\boxed{\langle a, b \rangle = 0}$ or $\boxed{a^T b = 0}$

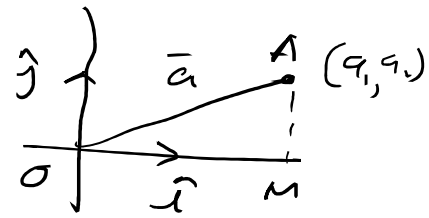
$\hat{j} = (0, 1)$

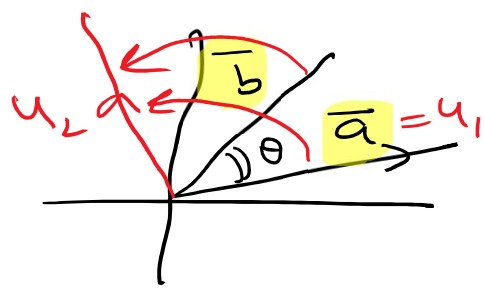
$\hat{i} = (1, 0)$

$OM = \bar{a} \cdot \hat{i} = a_1$

$MA = \bar{a} \cdot \hat{j} = a_2$

$\langle \hat{i}, \hat{j} \rangle = \hat{i} \cdot \hat{j} = 0$ [orthogonal]





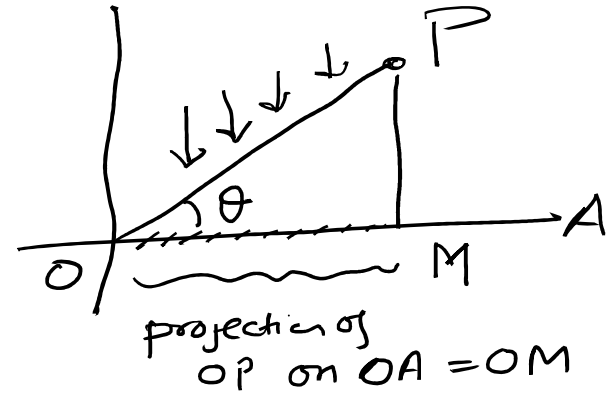
Q: Given a set of vectors $\{\bar{a}, \bar{b}\}$ can we find vectors $\{u_1, u_2\}$ s.t. $u_1 \perp u_2$.

$S = \{a, b\} \longrightarrow S_1 = \{u_1, u_2\}$ orthogonal.

YES! We can use "Gram Schmidt Orthogonalization"

$$OM = OP\cos\theta$$

$$= \text{Projection of } OP \text{ on } OA$$

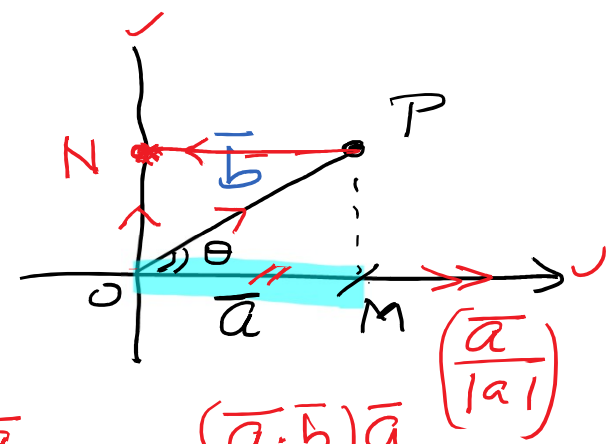


$$\cos\theta = \frac{\bar{a}\cdot\bar{b}}{|a|\,|b|}$$

$$\Rightarrow \underbrace{\underbrace{|b|}_{OP}\cos\theta}_{\text{Projection of } \bar{b} \text{ on } \bar{a}} = \overbrace{\left(\frac{\bar{a}}{|a|}\right)\cdot\bar{b}}^{\text{Scaler}}$$

unit vector

$$\text{Projection } OM = \left(\frac{\bar{a}}{|a|}\right)\cdot\bar{b}$$



$$\text{Vector in the direction of } OM = \underbrace{\left[\frac{\bar{a}\cdot\bar{b}}{|a|}\right]}_{\text{scaler}}\frac{\bar{a}}{|a|} = \frac{(\bar{a}\cdot\bar{b})\bar{a}}{|a|^2}$$

$$= \vec{\alpha}$$

Now $\overrightarrow{ON} = \vec{b} + \overrightarrow{PN}$

$\cancel{\overrightarrow{PN}}$

$$\overrightarrow{ON} = \vec{b} - \overrightarrow{OM}$$

$$\overrightarrow{ON} = \vec{b} - \underbrace{\frac{\vec{a}\cdot\vec{b}}{|a|^2}\,\vec{a}}$$

$$\overrightarrow{ON}\cdot\frac{a}{|a|} = b\cdot\frac{a}{|a|} - \frac{a\cdot b}{|a|^2}\,\frac{a\cdot a}{|a|} = \frac{b\cdot a}{|a|} - \frac{a\cdot b}{|a|} = 0$$

We showed $\underline{ON \perp OM}$

$$\left\{\overrightarrow{OM}, \frac{\overrightarrow{ON}}{\overrightarrow{ON}}\right\} \equiv \{u_1, u_2\}$$

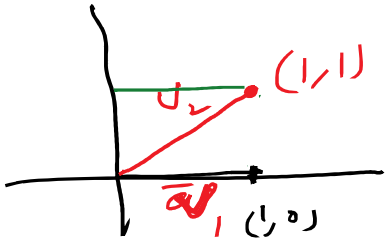
s.t. $u_1 \perp u_2$

$\|u_1\| = 1 \qquad \|u_2\| = 1$ ✓

Have a look at this example in 2D. ($\mathbb{R}^2$)

<u>Suppose</u> $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\quad |v_1| = 1$

$v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$



<u>Check</u> if $v_1 \perp v_2$? $v_1 \cdot v_2 = v_1^T \cdot v_2 = (1,0)(1,1) = \underline{1} \neq 0$

<u>NO</u>

We shall find $\{u_1, u_2\}$ from $\{v_1, v_2\}$ s.t. $u_1 \perp u_2$.

$u_1 = v_1$

$$u_2 = v_2 - \overbrace{\frac{v_1 \cdot v_2}{|v_1|} \cdot \frac{v_1}{|v_1|}}^{\text{projection of } v_2 \text{ on } v_1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{1}{1^2}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \underline{\underline{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}}$$

Gram Schmidt Orthogonalization ($\mathbb{R}^n$)

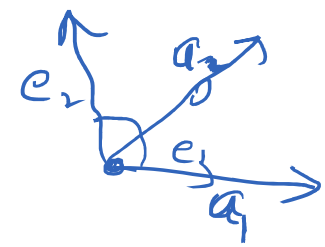
$$a_i = \begin{bmatrix} a_{i1} \\ a_{i2} \\ \vdots \\ a_{in} \end{bmatrix}$$

vector $S = \{a_1, a_2, a_3, \ldots, a_n\}$

$\longrightarrow S_1 = \{u_1, u_2 \ldots u_n\}$ s.t. S_1 is orthogonal. i.e. $(u_i \perp u_j \ \forall i, j)$

$$e_1 = \frac{u_1}{\|u_1\|}$$ unit vector

Process $\longrightarrow u_1 = a_1$



$$\longrightarrow u_2 = a_2 - \underbrace{\langle a_2, e_1 \rangle}_{\text{project.}} \underbrace{e_1}_{\text{unit vector}}$$

$$\boxed{e_2 = \frac{u_2}{\|u_2\|}}$$

$$\langle a_2, e_1 \rangle = a_2^T e_1 = a_2^T \frac{u_1}{\|u_1\|} = \underbrace{\frac{a_2^T u_1}{\|u_1\|}}_{\text{projection of } a_2 \text{ on } u_1}$$

Claim $u_2 \perp u_1$ i.e. $\langle u_2, u_1 \rangle = 0$

equivalent $\langle u_2, e_1 \rangle = 0$

LHS

$$\langle u_2, e_1 \rangle = u_2^T e_1$$

$$= (a_2 - \underbrace{\langle a_2, e_1 \rangle}_{\text{scalar}} e_1)^T e_1$$

$$= (a_2^T - \langle a_2, e_1 \rangle e_1^T) e_1$$

$$= a_2^T e_1 - \langle a_2, e_1 \rangle e_1^T e_1$$

$$= \langle a_2, e_1 \rangle - \langle a_2, e_1 \rangle = 0$$

$$= \text{RHS}$$

$[(A-B)^T = A^T - B^T$

$e_1 \to$ unit vector

$\because e_1^T e_1 = 1$

because

$e_1^T e_1 = \|e_1\|^2$

$= 1$

By mistake I deleted slides of the class and saved file. Therefore I rewrote it. It may be different here & there.

Similarly,

$$\longrightarrow \quad u_3 = a_3 - \langle a_2, e_1 \rangle e_1 - \langle a_2, e_2 \rangle e_2$$

and the unit vector $\boxed{e_3 = \dfrac{u_3}{\|u_3\|}}$

<u>Claim</u> $u_3 \perp e_1$ & $u_3 \perp e_2$ (or $u_3 \perp u_1$ and $u_3 \perp u_2$)

<u>you can verify yourself.</u>

Continuing in similar manner

$$u_n = a_n - \langle a_n, e_1 \rangle e_1 - \langle a_n, e_2 \rangle e_2 - \cdots - \langle a_n, e_{n-1} \rangle e_{n-1}$$

and $\boxed{e_n = \dfrac{u_n}{\|u_n\|}}$

Therefore, $S = \{a_1, a_2, \ldots a_n\} \xrightarrow{\text{orthogonal}} \{u_1, u_2, \ldots u_n\}$

$$\downarrow$$

$$\{\langle u_i, u_j \rangle = 0 \quad i \neq j\}$$

In the process we get $\{e_1, e_2 \ldots e_n\}$

s.t. $\langle e_i, e_j \rangle = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$.